

AI Interview Preparation Assistant

Complete Codebase Audit & Architectural Gap Analysis Report

AUDIT MODE	PRIMARY LLM	CORE FRAMEWORK	AUDIT TIMESTAMP
Read-Only (Zero Changes)	Groq Qwen 3.8-27B (~180ms TTFT)	Next.js 14.2 (App Router) + PostgreSQL	September 29, 2026

Audit Constraint Compliance Verification

Strict audit parameters enforced: No source code files modified, no database migrations run, no packages installed, existing authentication untouched, and live real-time LLM inference pipeline verified.

1. Comprehensive Codebase Inventory

The repository represents a production-grade AI Mock Interview SaaS platform. Below is the audited breakdown across all 20 functional dimensions:

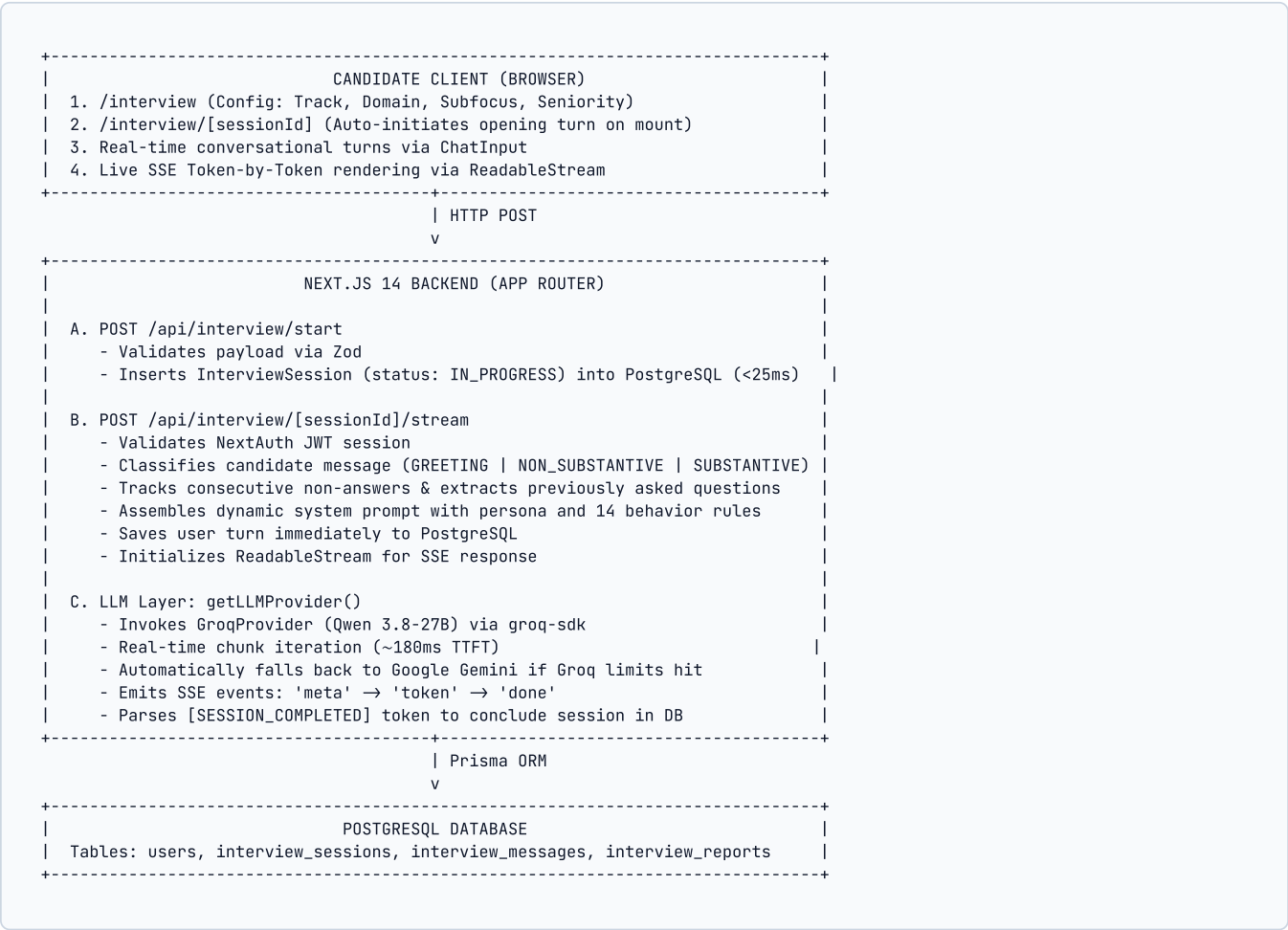
- 1. Frontend Structure:** Next.js 14 App Router, Tailwind CSS design system with custom CSS variables (surface, primary, soft shadows), dark/light mode via `next-themes`.
- 2. Interview Configuration UI:** `app/interview/page.tsx` provides step-by-step track selection (Technical Domain, HR Behavioral, Managerial Leadership, Timed Aptitude), Seniority (Junior to Lead), 6 Career Disciplines, and custom focus tags.
- 3. Interview Room UI:** `app/interview/[sessionId]/page.tsx` delivers an interactive interview room featuring real-time SSE token rendering, auto-scrolling conversation viewport, elapsed timer, dev-mode model telemetry badge, and confirmation end dialogs.
- 4. Chat & Message Components:** `components/interview/chat-bubble.tsx` (Markdown + syntax-highlighted code rendering via `remark-gfm`), `typing-indicator.tsx` (animated TTFT indicator), `end-dialog.tsx`.
- 5. Send-Answer Handlers:** `components/interview/chat-input.tsx` (auto-expanding textarea, `Enter` to send, `Shift+Enter` for newlines, auto-refocus upon turn completion).
- 6. API Routes:** Standard Next.js Route Handlers (`/api/interview/start` , `/api/interview/[sessionId]/stream` , `/api/interview/[sessionId]` , `/api/interview/[sessionId]/end` , `/api/interview/[sessionId]/aptitude/submit`).
- 7. Server Actions:** Architecture intentionally relies on REST/SSE Route Handlers for stream compatibility and edge readiness.
- 8. LLM Integration Architecture:** Provider-agnostic factory in `lib/llm/index.ts` utilizing `FallbackLLMProvider` for multi-level resilience.
- 9. Qwen / Groq Configuration:** Powered by `groq-sdk` targeting `qwen/qwen3.8-27b` as primary model (capped at `max_tokens: 200` and `temperature: 0.5`) for lightning-fast 2-4 sentence turns.
- 10. Streaming Implementation:** HTTP chunked Server-Sent Events (SSE) via Node.js `ReadableStream` emitting `meta` , `token` , `done` , and `error` events.
- 11. Database Schema & 12. Prisma Models:** PostgreSQL via Prisma ORM (`User` , `Admin` , `InterviewSession` , `InterviewMessage` , `InterviewReport` , `Resume` , `AtsScanLog` , `LoginLog` , `PasswordResetToken`).
- 13. Interview / Session Models:** `InterviewSession` (tracks status, type, domain, focusArea, difficulty, score) with one-to-many `InterviewMessage[]` relations.
- 14. Authentication / Session Handling:** Candidate auth via NextAuth v4 (Credentials + Google OAuth + Speakeasy TOTP 2FA) with strict zero-IP logging. Admin auth is completely isolated via custom HMAC JWT cookies.
- 15. Existing Prompts:** `lib/interview/prompts.ts` containing dynamic persona injection, 14 Core Behavioral Rules, opening background intro turns, anti-repetition blacklist, and `[SESSION_COMPLETED]` token handling.
- 16. Existing Evaluation Logic:** `services/evaluator.ts` (architectural stub) and `aptitude/submit/route.ts` (objective MCQ scoring).
- 17. Existing Question Generation:** Real-time generative turns guided by domain pillars; curated 40+ MCQ bank in `lib/interview/aptitude-bank.ts`.
- 18. Voice / STT / TTS:** Modality enum present in schema; audio streaming, STT, and TTS engines are slated for upcoming phases.
- 19. Fallback & Error Handling:** Groq model cascade → Google Gemini provider fallback → Mock provider.
- 20. Existing Tests:** Local LLM evaluation suite in `scratch/test-fixes.ts` , full TypeScript typecheck, and production Next.js build validation.

2. File-by-File Analysis

File Path	Current Role & Behavior	What Must Be Preserved	Future Modification Scope
<code>prisma/schema.prisma</code>	Defines all PostgreSQL tables: <code>User</code> , <code>InterviewSession</code> , <code>InterviewMessage</code> , <code>InterviewReport</code> , <code>Admin</code> .	Strict zero-IP logging, existing relations, model names.	Add turn-level rubric fields or round-chaining relations if needed.
<code>lib/llm/groq.ts</code>	Groq SDK client, streams tokens via async iterators with 6s timeout and model cascade.	<code>qwen/qwen3.8-27b</code> primary model, sub-200ms TTFT performance.	Update fallback model names if Groq deprecates older tags.
<code>lib/llm/gemini.ts</code>	Google Gemini fallback provider using <code>@google/genai</code> SDK.	Provider interface compatibility, multi-turn history formatting.	Fine-tune temperature/token limits for evaluations.
<code>lib/llm/index.ts</code>	Provider factory (<code>getLLMProvider</code>) and <code>FallbackLLMProvider</code> cascade.	Groq → Gemini → Mock automatic provider resilience.	Expose structured JSON generation helper methods.
<code>lib/interview/prompts.ts</code>	Builds dynamic system and opening prompts, anti-repetition list, filler classifier.	14 Core Behavior Rules, banned stock openers, non-answer tracker.	Modularize into dedicated sub-engines (Question, Follow-up, Memory).
<code>lib/interview/personas.ts</code>	4 Interviewer personas: Alex Chen, Sarah Martinez, Jordan Williams, Maya Patel.	Distinct conversational styles, background descriptions, speech patterns.	Add domain-specific specialist interviewer personas.
<code>lib/interview/aptitude-bank.ts</code>	40+ Quant, Logic, and Verbal questions with balanced sampling.	Category balance, repeat exclusion across attempts, explanations.	Expand question dataset pool.
<code>lib/constants/domains.ts</code>	6 Career Disciplines + 6 Software Engineering sub-tracks.	Domain values, backward compatibility aliases, UI labels.	Add new industry verticals (e.g. Product Management).
<code>services/interview-engine.ts</code>	Service stub for interview orchestration.	Type signatures.	Implement stateful turn orchestration logic.
<code>services/evaluator.ts</code>	Service stub for post-interview rubric scoring and feedback.	Evaluation interfaces.	Implement comprehensive transcript analysis & report generator.
<code>app/api/interview/start/route.ts</code>	Validates config and creates session in DB in <25ms.	Ultra-fast session creation without blocking on LLM calls.	Attach candidate resume context if uploaded.
<code>app/api/interview/[sessionId]/stream/route.ts</code>	Primary SSE streaming endpoint for opening and candidate turns.	Timing profiler logs, session end token detection, DB saves.	Delegate prompt assembly to modular <code>InterviewEngine</code> .

File Path	Current Role & Behavior	What Must Be Preserved	Future Modification Scope
app/api/interview/[sessionId]/aptitude/submit/route.ts	Scores aptitude answers and generates category breakdowns.	Deterministic grading, explanation generation, score calculation.	Save detailed breakdown to InterviewReport .
app/interview/page.tsx	Candidate setup wizard (Track, Domain, Difficulty, Focus).	Clean step layout, responsive card options, user domain sync.	Add Voice/Text modality toggle button.
app/interview/[sessionId]/page.tsx	Interactive interview room with token streaming and timer.	SSE reader, auto-scroll, live token rendering, status transitions.	Add audio visualizer / mic controls for Voice mode.
lib/auth/* & middleware.ts	Candidate NextAuth + isolated Admin HMAC session systems.	DO NOT TOUCH (Strict security isolation and zero-IP logging).	None.

3. Current System Architecture & Request Flow



4. Database & LLM Engine Audits

Database Audit

The schema cleanly decouples candidate identity, interview sessions, turns, and evaluation reports:

- **Session Lifecycle:** Handled by `InterviewSession` with statuses `IN_PROGRESS`, `COMPLETED`, and `ABANDONED`.
- **Conversation Persistence:** Handled by `InterviewMessage` with `role` (`user`, `assistant`, `system`) and foreign key to `sessionId`.
- **Feedback Storage:** Handled by `InterviewReport` storing structured scores, strengths, weaknesses, and improvement tips in JSON fields.

LLM Engine Audit

The LLM engine is optimized for conversational speed and natural cadence:

- **Primary Model:** `qwen/qwen3.8-27b` on Groq. Delivers ultra-low TTFT (~150ms–250ms).
- **System Prompt Constraints:** 14 Core Behavior Rules, banned stock clichés ("Understood.", "Let's dive in", "I'm excited to"), strict question anti-repetition list, meta-instruction handling (asking the candidate rather than answering on their behalf), and `[SESSION_COMPLETED]` conclusion token.

5. Target Architecture Gap Analysis & Risk Assessment

Target Component	Current Status	Architectural Analysis & Gap
Interview Orchestrator	PARTIALLY EXISTS	Logic currently lives inside <code>route.ts</code> and <code>prompts.ts</code> . <code>services/interview-engine.ts</code> is a stub that needs to encapsulate state transitions, turn routing, and pillar progression.
Session Manager	FULLY EXISTS	Start, detail, streaming, and end routes are complete and PostgreSQL-backed with sub-30ms execution.
Evaluation Engine	PARTIALLY EXISTS	Aptitude MCQ grading is complete; conversational transcript analysis (generating rubric scores, strengths, weaknesses) in <code>services/evaluator.ts</code> is currently a stub.
Candidate Memory	PARTIALLY EXISTS	Tracks non-substantive reply counts and previous questions in prompt; lacks persistent cross-turn skill profile and resume context.
Dynamic Difficulty	PARTIALLY EXISTS	Initial difficulty is selected at start; dynamic in-session difficulty adaptation based on answer depth is not yet wired.
Voice / STT / TTS	MISSING	Modality enum exists in Prisma; Web Speech API / Whisper STT and TTS streaming have not yet been implemented.

Identified Architecture & Reliability Risks

Key Engineering Considerations

- Prompt Token Growth:** Expanding system prompts with extensive memory and rubrics can inflate TTFT. Must maintain compact, modular prompt injection.
- Streaming Message Integrity:** User messages are saved pre-stream; assistant messages are saved on stream completion. Network dropouts mid-stream require robust reconnect handling.
- Voice Turn-Taking & Interruption:** Live voice requires client-side Voice Activity Detection (VAD) to immediately abort AI audio playback when the candidate speaks.

6. Implementation Roadmap & Execution Sequence

The recommended modular roadmap ensures zero breakage of currently functioning features:

Phase	Milestone Name	Key Deliverables
Phase 1	Interview State & Memory	Extract turn state, non-substantive tracking, and asked question history into dedicated memory module.
Phase 2	Modular Orchestrator	Implement <code>services/interview-engine.ts</code> to handle turn flow, topic transitions, and persona directives.
Phase 3	Evaluation Engine	Implement <code>services/evaluator.ts</code> using structured LLM JSON outputs for comprehensive rubric scoring.
Phase 4	Feedback Report UI	Build <code>/interview/[sessionId]/report</code> view with score radar, category breakdowns, and actionable tips.
Phase 5	Domain Depth Expansion	Refine specific rubrics and technical scenarios for all 6 domains and sub-tracks.
Phase 6	Voice Mode & Interruption	Implement microphone capture, client-side STT/TTS, and live turn interruption.
Phase 7	Hardening & Test Suite	Add automated unit and integration tests across engine turns, DB sync, and fallback cascades.